Parniyan Malekzadeh

Computer Engineering Graduate

parnian.malekzadeh1@gmail.com | parmalvar.github.io | parmalvar | parniyanmalekzadeh

Summary

Computer Engineering BSc graduate from Isfahan University of Technology with an 18.05/20.0 GPA, ranked in the top 10% of the 2021 Computer Engineering cohort. Currently conducting supervised research on concept drift in machine-learning-based malware detection. Strong academic background in secure computing, with experience spanning applied cryptography, secure software development, network security, and machine learning.

Education

Isfahan University of Technology — **2021 – 2026**
Bachelor of Science in Computer Engineering – Secure Computing — *Isfahan, Iran*

- **GPA:** 18.05/20.0; **Major GPA:** 4.0/4.0
- Ranked in the **Top 10%** of the 2021 Computer Engineering cohort.
- **BSc Project:** Design and Development of a Platform to Connect Advertisers and Promoters
- **Supervisor:** Dr. Hossein Falsafain

National Organization for Development of Exceptional Talents (Sampad) — **2018 – 2021**
High School Diploma in Mathematics and Physics — *Isfahan, Iran*

- **GPA:** 19.8/20.0

Research Experience

Concept Drift in Malware Detection — **June 2026 – Present**

- Conducting research under the supervision of **Dr. Mohammad Dakhilalian** and **Dr. Mojtaba Khalili** on concept drift in machine-learning-based malware detection, with a focus on the degradation of detection models as malware characteristics and data distributions evolve over time.
- Reviewing and presenting research on **malware detection and drift-aware learning**, covering static and deep-learning-based detection, temporal distribution shifts, drift detection, and model adaptation.
- Studying approaches for addressing evolving malware distributions, including **active and semi-supervised learning, pseudo-labeling/self-training, explainable drift detection, rule-based methods, and few-shot adaptation**, across Android and Windows malware settings.

Research Interests

- Applied Cryptography
- Network Security & SDN
- Secure Software Development
- Malware Detection & Concept Drift
- Game Theory in Security
- Forensics & Incident Response

Experience

Isfahan University of Technology — **2024 – 2026**
Teaching Assistant — *Isfahan, Iran*

- **Courses:** Secure Computing (**Head TA**), Secure Software Development (**Head TA**), Fundamentals of Cryptography, Artificial Intelligence, Automata Theory, IT Fundamentals, Database Design, and Applied Linear Algebra.
- Designed and evaluated a multi-stage **RSA implementation and security-analysis project**, covering modular exponentiation, Miller–Rabin primality testing, key generation, CRT-based decryption optimization and benchmarking, **Simple Power Analysis (SPA)**, private-key recovery from power traces, and constant-time countermeasures.
- Designed and graded assignments and projects and delivered problem-solving or solution sessions across courses in artificial intelligence, automata theory, databases, information technology, and applied linear algebra.

Kasra Engineering Co. — **Jul. 2024 – Sep. 2024**
Software Engineering Intern — *Isfahan, Iran*

- Developed a **Kotlin/MVVM attendance-management module**, integrating server APIs using Retrofit and Gson and managing reactive application state with ViewModel, LiveData, and Coroutines.
- Implemented **offline data persistence and synchronization** using Room; optimized database queries, network caching, and memory usage, including diagnosing and resolving memory leaks with Android Profiler and LeakCanary.
- Implemented **unit, UI, and integration tests** using JUnit, Mockito, and Espresso, and worked within Git-based code-review and Jenkins CI/CD workflows.

Honors & Awards

Direct Admission to the MSc Program **2025**
Isfahan University of Technology *Isfahan, Iran*

- Granted based on outstanding undergraduate academic performance and ranking among the top Computer Engineering students.

Membership in the Brilliant Talents and Exceptional Students Program **2025**
Isfahan University of Technology *Isfahan, Iran*

- Recognized for outstanding academic performance.

National University Entrance Examination (Konkur) **2021**
Mathematics and Physics *Iran*

- Ranked 591 among 160,000+ participants (Top 0.3%).

Projects

Adversarial Robustness of Flow-Based Network Intrusion Detection **2026**

- Implemented and evaluated **FGSM and PGD adversarial attacks** against a CIC-IDS2017 flow-based MLP NIDS, including a padding/delay mutability mask that freezes immutable features and restricts selected traffic statistics to increase-only changes.
- Unconstrained PGD reached an **82% attack success rate (ASR)**; the mutability mask reduced ASR to **46%**, constrained examples transferred to a random forest at **61% ASR**, and adversarial training reduced constrained ASR to **8%** with clean F1 of 0.957.

RSA-CRT Side-Channel Analysis **2026**

- Developed a reproducible **simulated Simple Power Analysis (SPA)** framework for RSA-CRT, implementing square-and-multiply leakage, d_p recovery, and modulus factorization via GCD.
- Evaluated countermeasures including message/exponent blinding, dummy multiplication, and the Montgomery ladder; unprotected RSA-CRT yielded **100% exact d_p recovery** in 1024-bit trials, while ladder and dummy multiplication reduced exact recovery to **0%**.

Game-Theoretic IDS Sampling **2025 – 2026**

- Extended a theoretical IDS-sampling model into a reproducible simulation framework for allocating limited sampling budgets using **max-flow/min-cut**, with uniform and randomized baselines.
- Evaluated multiple topologies and attacker strategies using seeded Monte Carlo experiments; on a 3×3 grid at $B_s/\mathrm{MF} \approx 0.30$, min-cut allocation achieved detection probability **0.300** versus **0.161** for uniform sampling (**87% relative improvement**).

Network Security CTF **2025**

- Completed a closed-lab multi-host CTF involving **PHP type juggling, Zimbra XXE, VPN/SSH access, lateral movement, and SSH tunneling**, and used **Wiener's attack** against a weak RSA key to reach an additional host.

CERT and Incident Response Research **2024**

- Conducted a literature-based study of **CERT/CSIRT structures and the NIST incident-response lifecycle**, including coordination, containment, recovery, evidence handling, and a published financial-sector intrusion case study.

Selected Coursework

Security: Secure Software Development (20.00), Fundamentals of Cryptography (20.00), Network Security Lab (20.00), Fundamentals of Secure Computing (18.20)
Systems & Software: Computer Architecture & Organization (19.00), Computer Networks (18.60), Databases I (19.06), Software Testing (18.43), Operating Systems I (17.90), Software Engineering I (18.00), Fundamentals of Information Technology (18.30)
AI & Mathematical Foundations: Game Theory (18.30), Artificial Intelligence (18.14), Applied Linear Algebra (18.10), Theory of Formal Languages (18.19)

Technical Skills

Programming Languages: Python, C/C++, Java, Kotlin, SQL, Bash, Assembly
ML & Scientific Computing: NumPy, pandas, scikit-learn, NetworkX, Matplotlib
Security: Applied Cryptography, Side-Channel Analysis (SPA), Network Intrusion Detection, Adversarial Machine Learning, Secure Software Development
Security & Networking Tools: Wireshark, Scapy, Nmap, Burp Suite, Metasploit, Mininet, Open vSwitch, OpenSSL, PyCryptodome
Development & Platforms: Git, Linux (Ubuntu/Kali), Docker

Languages

English: IELTS 7.5 overall (L 7.5, R 7.0, W 7.0, S 7.5), Dec. 2025
Persian: Native